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Marketing Campaign Analysis

Overview

This analysis examined customer demographics, purchasing behavior, product spending, campaign response, and complaints in the 2012-14 marketing campaign dataset. The dataset was cleaned by removing certain unusual marital-status and age records, and imputing missing income values using average income within education and marital-status groups.

Feature-engineered variables included total children, age, total spending, and total purchases. Outliers in income and total spending were addressed via winsorization.

Key Findings

1. Purchasing and spending

Total purchases were calculated from web, catalog, and store purchases. Purchases via deals were not treated as a separate channel because they can overlap with the other three purchasing channels. Product spending analysis showed that wine and meat generated the highest revenue, while fruits generated the lowest revenue.

2. Hypothesis testing

- Older vs. younger customers: The analysis found sufficient evidence that older adults make more in-store purchases than younger adults.
- Customers with vs. without children: There was insufficient evidence that customers with children make more online purchases.
- Physical stores vs. alternative channels: The analysis found sufficient evidence that in-store purchases are lower than the same customers' purchases through alternative channels. This suggests that alternative distribution channels may be associated with lower purchase activity at physical stores.
- United States vs. non-U.S. customers: There was insufficient evidence to conclude that the U.S. has a higher average purchase volume per customer.

3. Campaign acceptance

Although the oldest customers had the highest response rate, the visualization did not show a clear consistent relationship between age and acceptance. The country with the highest number of customers accepting the last campaign was Spain (SP).

4. Children and expenditure

Average total expenditure was highest among customers with no children and generally decreased as the number of children increased.

5. Complaints and education

Customers with a Graduation-level education accounted for the largest number of complaints among the education groups shown. Customers with a Basic education level had zero complaints and therefore did not appear in the complaint bar chart.

Business Recommendations

- Prioritize alternative sales channels, which showed higher average purchase counts than physical stores.
- Target older customers with in-store promotions, given their higher average in-store purchase activity.
- Avoid assuming customers with children are stronger online buyers, as the analysis found insufficient evidence for this.
- Investigate the observed difference in U.S. versus non-U.S. purchasing further, since it was not statistically significant.
- Consider age-specific campaign strategies, since acceptance rates varied across age groups without a clear consistent trend.
- Focus cross-selling and promotional strategies around wine and meat, which had the highest spending among the product categories.
- Analyze Spain's strong campaign response volume to identify effective campaign elements that could be tested in other markets.
- Monitor complaint rates by education level relative to the size of each education group before targeting any particular group for corrective action.